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Data-driven prediction of dissolved oxygen to identify fish kill factors: Case study in Redbank weir at Murrumbidgee River, Australia

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ABSTRACT

Dissolved oxygen (DO) is critical in aquatic ecosystem health, prompting the need for precise predictive models. This study introduces a novel Convolutional Neural Network-Gated Recurrent Unit (CNN-GRU) framework to predict DO levels at Redbank Weir on the Murrumbidgee River, Australia. Considering a definition of stratification that accounts for thermal stratification, alongside satellite data to obtain an estimation of chlorophyll content in waters. The models were trained using data from 2017 to 2024. The CNN-GRU model demonstrates superior performance, with validation metrics indicating a Mean Absolute Error (MAE) of 0.6157 and Root Mean Square Error (RMSE) of 0.7521 mg/L for daily predictions, and MAE = 0.6391 and RMSE = 0.6356 for hourly predictions. Furthermore, the explainable AI analysis reveals strong correlations between DO levels and surface water temperature, while stressing the significant impacts of water level and evaporation on overall model explainability. The analysis highlighted the importance of intricate interactions among environmental variables associated with low DO levels during fish kill events. These factors interact in complex ways: for example, higher temperatures reduce oxygen solubility while simultaneously accelerating microbial respiration, which further consumes DO. This research emphasises not only predictive accuracy but also the importance of explainability in machine learning models. Ultimately, providing stakeholders with insights for informed decision-making regarding aquatic ecosystem management and water quality regulation.

1. Introduction

Dissolved oxygen (DO) is a critical component that plays a significant role in biochemical processes within water bodies. It significantly affects the respiration rates of aquatic organisms, the decomposition processes of organic matter, the oxidation of pollutants, the thermal dynamics of water, the structural characteristics of habitats, and the intricacies of food webs (Boyd, 2020). To live and reproduce, fish and other aerobic aquatic animals need oxygen (Caldwell and Hinshaw, 1994). Various biotic and abiotic factors influence DO levels. Among these factors, water temperature is a key determinant (Zhong et al., 2021), and thermal stratification is a major driver of hypoxia in many aquatic systems, as it creates a barrier to vertical mixing. Thermal stratification prevents oxygen-rich surface waters from replenishing oxygen in deeper layers, leading to a gradual depletion of DO in the hypolimnion (Queste et al., 2013). Recognising the interconnectedness of these factors can guide effective interventions to mitigate hypoxia and promote healthier aquatic environments.

Since phytoplankton and algae produce oxygen through photosynthesis, fluctuations in chlorophyll concentrations can influence DO levels. High chlorophyll concentrations typically correlate with nutrient enrichment, which can influence DO levels by increasing biological

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activity and organic matter decomposition (Coffin et al., 2021). The temperature sensitivity of chlorophyll-a was also demonstrated by Zhu et al. (2022). During algal blooms, elevated chlorophyll can lead to increased oxygen production during daylight hours; however, when these blooms decay, microbial decomposition consumes large amounts of oxygen, often resulting in hypoxic conditions (Holmer, 1999). Moreover, seasonal variations, nutrient inputs, and environmental factors modulate this relationship, underscoring the need to study their interplay for effective water management and pollution control strategies (Wang et al., 2002). By elucidating the interaction between chlorophyll and DO, researchers and water managers can better predict, monitor, and mitigate eutrophication and hypoxia, thereby ensuring the sustainability of aquatic ecosystems.

The primary focus of this work is to develop data-driven methods to identify the factors driving fish kills. Fish kills are events characterised by the sudden or gradual death of large numbers of fish within a water body. These incidents are often symptomatic of underlying environmental stressors, with low DO being a primary cause (Chen et al., 2020). Fish kills can result from natural phenomena, such as temperature variations and water stratification, or from anthropogenic activities, including nutrient pollution, eutrophication, and habitat modification (Mohd-Din et al., 2020). In Australia, fish kills have been common in the Murray-Darling Basin. In the Menindee region and the Murrumbidgee River, hundreds of thousands of fish have been killed in recent years (Murray–Darling Basin Authority, 2019). According to reports, thermal stratification is rather prevalent in Australia's lowland rivers and lakes (Australian Academy of Science, 2019). Baldwin (2019) reports that the 2019 fish kill at Redbank Weir in the Murrumbidgee River was caused by pool mixing and subsequent destratification. During these events, DO levels have dropped dramatically from a high of around 10 mg/L to 1.5 mg/L within a few hours or days. The consequences of fish kills extend beyond ecological impacts, affecting local economies, fisheries, and community livelihoods.

In recent years, increasing interest in DO modelling and forecasting has driven extensive research. However, the multitude of factors affecting DO variability—ranging from meteorological conditions to complex biological interactions—remains dynamic and inadequately understood. (Zhi et al., 2021). Efforts in modelling oxygen dynamics have employed process-based models grounded in physical, chemical, and biological principles. These include: (i) steady-state mass balance models such as the Streeter–Phelps equation; (ii) dynamic models using differential equations (e.g., RWQM, QUAL2K, WASP); (iii) hydrodynamic–water quality models that integrate flow and mixing with water quality (e.g., CE-QUAL-W2, EFDC); and (iv) eutrophication models that couple nutrients, algal growth, and DO dynamics (e.g., GEMS, PCLake). These models range from one-dimensional to advanced three-dimensional hydrodynamic and biochemical processes, combining numerous parameters and complex nonlinear interactions. As such, these models require a high level of expertise to develop, maintain, and use, and are generally not easily accessible (Joehnk et al., 2020).

More recently, prediction models for DO concentration have been developed using data-driven techniques. Data-driven models for DO rely on statistical/empirical relationships between historical data and DO measurements. Banerjee et al. (2019) employed artificial neural networks to predict DO levels based on environmental factors. Overall, neural network modelling proves to be a powerful tool for understanding and predicting ecosystem dynamics, offering advantages over traditional approaches. Long Short-Term Memory (LSTM) networks have been extensively utilised for DO prediction due to their ability to handle sequential data effectively. Anuar et al. (2021) and Hao (2024) both employed LSTM models to predict DO, emphasising their ability to model temporal dependencies effectively. Garabaghi (2025) predicted DO levels using a Deep Neural Network (DNN) model enhanced by Bayesian optimisation for hyperparameter tuning. The authors demonstrate that the Bayesian-optimised model significantly outperforms a baseline model with default settings, highlighting the effectiveness of Bayesian methods in improving model accuracy and generalization. However, the computational complexity of Bayesian Optimisation can lead to resource-intensive models. These studies demonstrate the potential of data-driven techniques to understand and manage oxygen dynamics in aquatic systems.

Remote sensing technologies have been effectively integrated with machine learning models to estimate DO levels. Using Sentinel-2 images and an ensemble machine learning algorithm, Zhu et al. (2022) estimated the DO in Shenzhen Bay. With an inaccuracy of just 0.02%, the model demonstrated good accuracy in estimating DO. Beal et al. (2024) employed remote sensing combined with various machine learning algorithms to estimate DO and algae pigments in inland lakes, achieving significant correlations and highlighting the importance of algae dynamics in oxygen variability. As technological advancements continue, interdisciplinary approaches combining satellite data will be crucial for advancing our understanding of the intricate relationships among different factors and their effects on DO in aquatic systems.

Explainable AI (xAI) in water-quality parameter analysis offers a multifaceted approach to leveraging machine learning and deep learning for water monitoring and management. Zhou and Zhang (2023) proposed an interpretable deep learning model integrating SHAP with LSTM to predict DO levels in karst spring flow. It highlights the complex hydrological processes affecting DO, emphasising the significant contributions of precipitation and discharge, while noting that higher water temperature and specific conductance negatively impact DO. However, the SHAP-LSTM model, while interpretable, may still struggle to fully capture the intricate physical processes that influence DO levels. In comparison, Mermer et al. (2025) present a comparative study of ensemble machine learning and explainable AI techniques for predicting harmful algal blooms in Lake Erie. The integration of SHAP analysis identifies key factors, such as particulate organic nitrogen, total phosphorus, and particulate organic carbon, as critical drivers of chlorophyll-a concentrations. However, the model was created using a limited number of stations because of missing data, and the training period was relatively brief.

Therefore, despite significant advancements, most existing DO prediction studies develop site-specific, black-box, point-forecast models with weak explainability, and limited evaluation of cross-site transfer, biological and ecological variables (such as phytoplankton blooms) and performance under extreme conditions, particularly in rivers and other underrepresented systems. Consequently, the objective of this study is to enhance the understanding of the most likely causes of fish deaths under specific drivers by predicting DO, using a deep learning approach, under persistent stratification conditions and considering phytoplankton blooms as a contributory factor in a case study at the Murrumbidgee River in the Murray-Darling Basin during the fish kill event on the 26th and 27th of January 2019.

This study makes three contributions. First, it assesses the impacts of persistent stratification and chlorophyll-a concentrations as key contributing factors to DO fluctuations, using the Murrumbidgee River in the Murray-Darling Basin, Australia, as a case study. Second, it introduces an AI-based CNN-GRU model to predict DO levels, offering enhanced predictive performance and representing a significant advancement in DO modelling. Finally, it compares the factors influencing both daily and hourly predictive models of DO, examining how these temporal scales affect model performance and their implications for understanding and mitigating fish kill events in the Murray-Darling Basin.

The study's remaining sections are organised as follows: Section 2 presents the study site, data collection and pre-processing, development of the proposed model, model performance, and explainability. Section 3 focuses on the prediction results and analysis of the objective and other models. Finally, the study's conclusions are presented in Section 4.

2. Materials and methods

2.1. Study site

This study focuses on the Redbank Weir, located on the Murrumbidgee River. The Murrumbidgee River, a significant tributary of the Murray River in the Murray–Darling Basin, is the third-longest in Australia, with approximately 1485 Km of river length and 4000 GL of annual stream flow (Murray–Darling Basin Authority, 2025). It passes through the regions of New South Wales (NSW) and the Australian Capital Territory, starting in the Australian alpine region (with over 1600 mm of annual average rainfall) until the semi-arid plains of western NSW (with around 350 mm of annual average rainfall) (Murray–Darling Basin Authority, 2025). Situated upstream of Balranald, the weir lies within the Lower Murrumbidgee wetlands in southern NSW. There are several weirs and a dam in the Murrumbidgee River: Balranald Weir, Redbank Weir, Maude Weir, Hay Weir, Gogeldrie Weir, Yanco Weir, Berembed Weir, and Burrinjuck Dam. Fig. 1 provides a visual representation of their geographic locations.

2.2. Data collection

Table 1 presents the predictive features used in this study. These features are the independent variables in the daily and hourly models, with data sets from January 2017 to December 2024. All of them keep a relationship with the DO in different ways.

Hydrological data, such as surface water temperature and water level, were obtained from WaterNSW (WaterNSW, 2025). Variations in hydrology, including water temperature and water level, significantly influence the concentration of DO in water bodies.

Meteorological data, such as rainfall, were obtained from the Australian Bureau of Meteorology (BoM) (Bureau of Meteorology, 2025), while data on air temperature, wind speed, shortwave irradiance, and relative humidity were sourced from Meteoblue (Meteoblue AG, 2025). The concentration of DO is influenced by multiple environmental factors, particularly meteorological conditions.

Gridded climate data, such as evaporation and air pressure, were

Table 1
Input features used to build the proposed models.

Acronym	Feature name	Source	Units	Period
MSWT	Mean surface water temperature	WaterNSW	°C	2015–2024
MNSWT	Minimum surface water temperature	WaterNSW	°C	2015–2024
MXSWT	Maximum surface water temperature	WaterNSW	°C	2015–2024
MWL	Mean water level	WaterNSW	m	2015–2024
MEVA	Mean evaporation	SILO	mm	2012–2024
MR	Mean rainfall	BoM	mm	1879–2024
MAP	Mean air pressure	SILO	hPa	2012–2024
MXAT	Maximum air temperature	Meteoblue	°C	2015–2024
MNAT	Minimum air temperature	Meteoblue	°C	2015–2024
MAT	Mean air temperature	Meteoblue	°C	2015–2024
MWS	Mean Wind speed	Meteoblue	m s^{-1}	2015–2024
MSR	Mean shortwave irradiance	Meteoblue	W m^{-2}	2015–2024
MXSR	Maximum solar irradiance	Meteoblue	W m^{-2}	2015–2024
MRH	Mean relative humidity	Meteoblue	%	2015–2024
MNRH	Minimum relative humidity	Meteoblue	%	2015–2024
MXRH	Maximum relative humidity	Meteoblue	%	2015–2024
MWTD	Mean water temperature difference (top - bottom)	LAKEoneD model	°C	2015–2024
MNDCI	Median normalised difference chlorophyll index	Sentinel-2	–	2017–2024
NM	Non-mixing criterion	Criterion	–	–

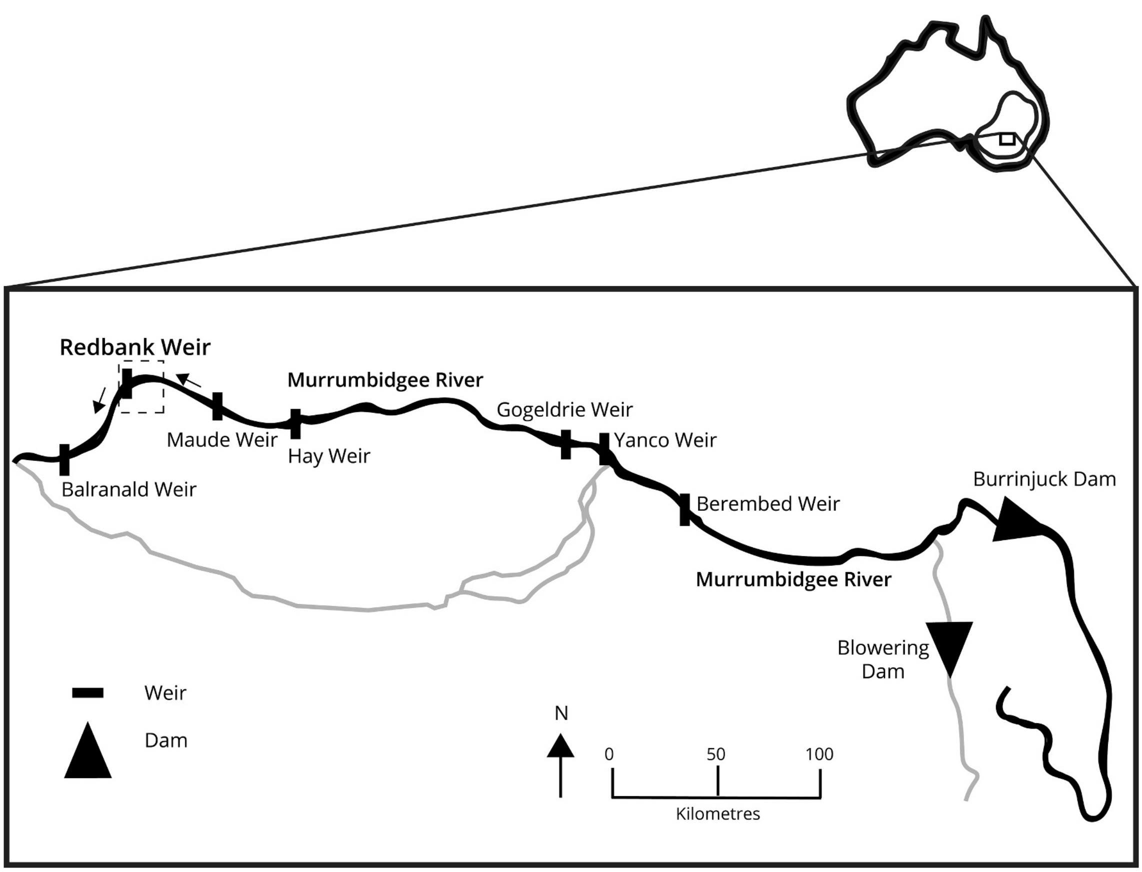


Fig. 1. Map of the study site, showing the Redbank Weir situated on the Murrumbidgee River, Region of New South Wales, Australia.

sourced from the Scientific Information for Land Owners (SILO) (Jeffrey et al., 2001). Gridded data is derived using various interpolation techniques that estimate values at unsampled locations based on observations from nearby locations.

Satellite data has become an invaluable resource for monitoring water quality parameters. Thus, in this study, satellite data were obtained from Sentinel-2 (Copernicus, 2025) to estimate the normalised difference chlorophyll index. Satellite-based chlorophyll estimation can detect fluctuations in chlorophyll concentration, which are key indicators of ecosystem dynamics, such as algal blooms or eutrophication.

Thermal stratification influences the distribution of DO. Thermal stratification occurs when water bodies develop layers of different temperatures, typically resulting in a warmer upper layer (epilimnion) and a cooler, denser lower layer (hypolimnion). In this study, thermal stratification is being considered through two sources: simulation data and non-mixing criterion. Simulation data on water temperature differences were derived from the LAKEoneD model (Joehnk et al., 2008; Joehnk and Umlauf, 2001). The LAKEoneD model was calibrated using data from the Redbank Weir on the Murrumbidgee River, which were provided by the authors of the study "Weir stratification and hypoxic water management - Murrumbidgee River 2019" (Baldwin, 2019). On the other hand, non-mixing refers to the phenomenon in which water layers of varying temperatures or densities remain distinct and do not intermingle. The non-mixing criterion is characterised by the continuous state of air temperature being lower than that of the water. This relationship leads to a systematic cooling effect at the water's surface, which prompts mixing processes to initiate. This criterion underscores the importance of monitoring atmospheric and water temperatures to anticipate mixing behaviour, which can significantly affect oxygen levels, and overall ecosystem health.

Alternatives beyond LAKEoneD range from enhanced 1-D column models like the General Lake Model (GLM), through 2D laterally resolved tools such as CE-QUAL-W2, to fully 3D hydrodynamic–ecological systems like Delft3D, and the appropriate choice depends on the spatial complexity, the processes of interest, and available data and computing power. However, in this study, only the LAKEoneD model was readily available.

The target variable (DO) and its descriptive statistics are summarised in Table 2.

2.3. Data pre-processing

Data pre-processing encompasses several techniques, including data cleaning and transformation. Part of the data cleaning process includes imputing missing data. The presence of missing data can severely limit the ability to perform accurate analyses, making imputation a critical step in data pre-processing.

The experiments conducted in this study evaluated the proposed procedure by Muzellec et al. (2020) on the dataset, addressing missing data under the Missing at Random (MAR) mechanism. The MAR mechanism refers to a situation in which the probability of data being missing depends on the observed data rather than on the missing data itself. This means that the values of other observed variables can explain the missingness. In this context, a subset with no missing values was first generated from the total dataset. Then, using the Muzellec procedure, a subset was randomly selected and evaluated using the previously mentioned imputation methods. The selection of the most suitable imputation method for these features was based on the precision of the corresponding metrics.

To address missing data (up to 40% for Median Normalised Difference Chlorophyll Index and $< 5\%$ for other predictors), we applied multiple imputation and interpolation methods to create imputed datasets. The selected imputation and interpolation methods are: Bayesian Ridge method (for Median Normalised Difference Chlorophyll Index), Spline interpolation method (for surface water temperature, maximum, minimum and mean air temperatures), Forward fill method (for water level and rainfall).

For instance, Fig. 2 illustrates plots of imputation methods evaluated for the Median Normalised Difference Chlorophyll Index at Redbank Weir on the Murrumbidgee River. Satellite data can often be incomplete due to various factors such as cloud cover, sensor malfunctions, or other environmental conditions. The imputation methods that have been used to handle missing data in satellite imagery are mean imputation, K-Nearest Neighbours (KNN), Random Forest (RF), Bayesian Ridge method, and Generative Adversarial Networks (GANs). The algorithm used for imputing GANs was GAIN (Yoon et al., 2018). After evaluating these methods, the highest r-value (0.8739) and the lowest RMSE (0.0937) were obtained using the Bayesian Ridge method.

Part of the data transformation process includes scaling. In this study, all input features were normalised to the range [0,1] using MinMaxScaler. This pre-processing step enables the learning algorithm to converge more efficiently and enhances the model's ability to identify the true patterns in the data.

2.4. Development of the proposed xAI model

Fig. 3 presents a flowchart outlining the primary stages of the model's development. It has seven components: data acquisition, data pre-processing, data splitting, model building, model training, model evaluation, and explainability. In data acquisition, we have as input data: meteorological and hydrodynamic factors, satellite data, the LAKEoneD model data, and the non-mixing criterion. In the pre-processing step, the data is cleaned and imputed using different imputation methods. The data is then split into two sets: a training dataset and a test dataset. The dataset has been split into training and test sets, maintaining the chronological sequence of observations. So, the test set contains data from subsequent time periods. The training dataset was used to train the model, while 25% of the data was reserved for testing and evaluation. In total, seven models were evaluated using a range of metrics. When training and evaluating our model, we explore different learning rates, architectures, and regularisation techniques to fine-tune the model. Following the evaluation, the best model was selected for both daily and hourly cases. Those two models were studied using explainable methods.

Neural networks have emerged as powerful tools in ecological modelling. Various neural network architectures have been employed in ecological modelling, each suited to specific types of data and prediction challenges (Abiodun et al., 2018). Thus, the most appropriate model is selected from a set of candidate models for a predictive task, such as simple Recurrent Neural Networks (RNNs) (Schmidt, 2019) (Fig. 4a), Long Short-Term Memory (LSTM) (Hochreiter and Schmidhuber, 1997) (Fig. 4b), Gated Recurrent Unit (GRU) (Cho et al., 2014) (Fig. 4c), Convolutional Neural Network (CNN) (O'shea and Nash, 2015) (Fig. 4d), Bidirectional LSTM (BiLSTM) (Fig. 4e), Bidirectional GRU (BiGRU)

Table 2
Descriptive statistics of the target variable (DO) from 01 January 2017 to 31 December 2024.

Variable	2017	2018	2019	2020	2021	2022	2023	2024
Mean DO (mg/L)	8.3506	8.8804	8.8522	8.8072	8.4676	7.1144	8.1096	8.9472
Minimum DO (mg/L)	4.2500	3.8810	0.0810	3.9270	3.6770	0.4280	3.3430	4.7000
Maximum DO (mg/L)	12.1140	15.7130	15.3080	12.5670	11.9980	10.8700	11.2130	12.9000
STD DO (mg/L)	2.0093	1.7318	2.0955	1.6240	1.8592	2.5392	1.6508	2.4210

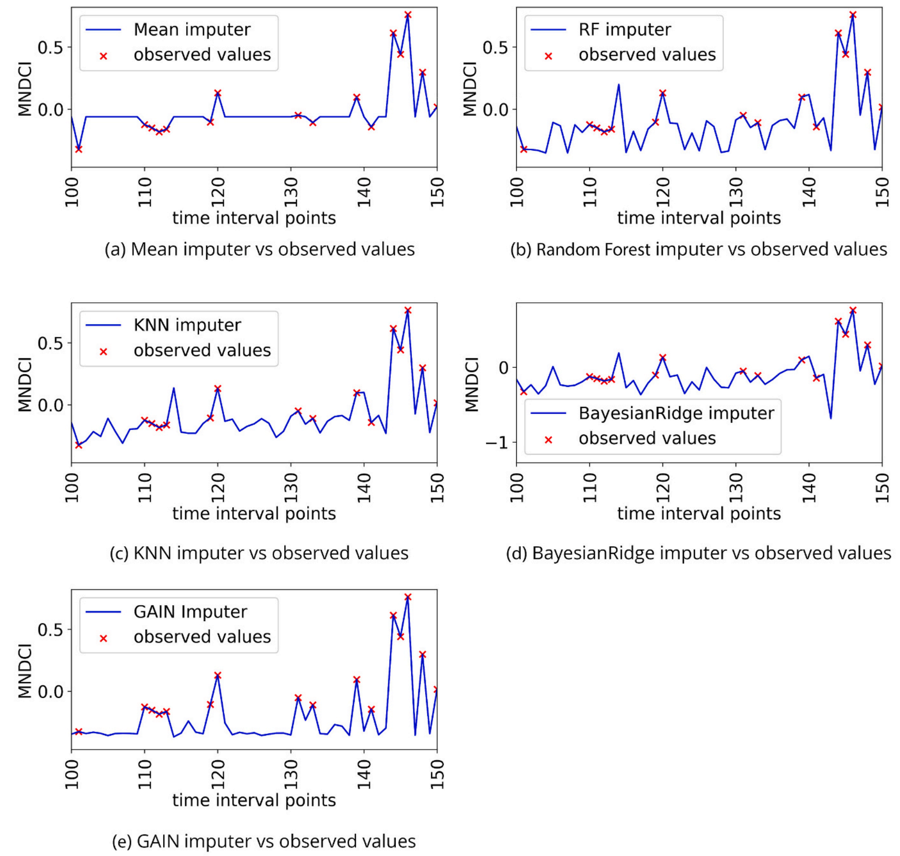


Fig. 2. Imputation methods applied for evaluating the Median Normalised Difference Chlorophyll Index (MNDCI) for the Redbank Weir at the Murrumbidgee River.

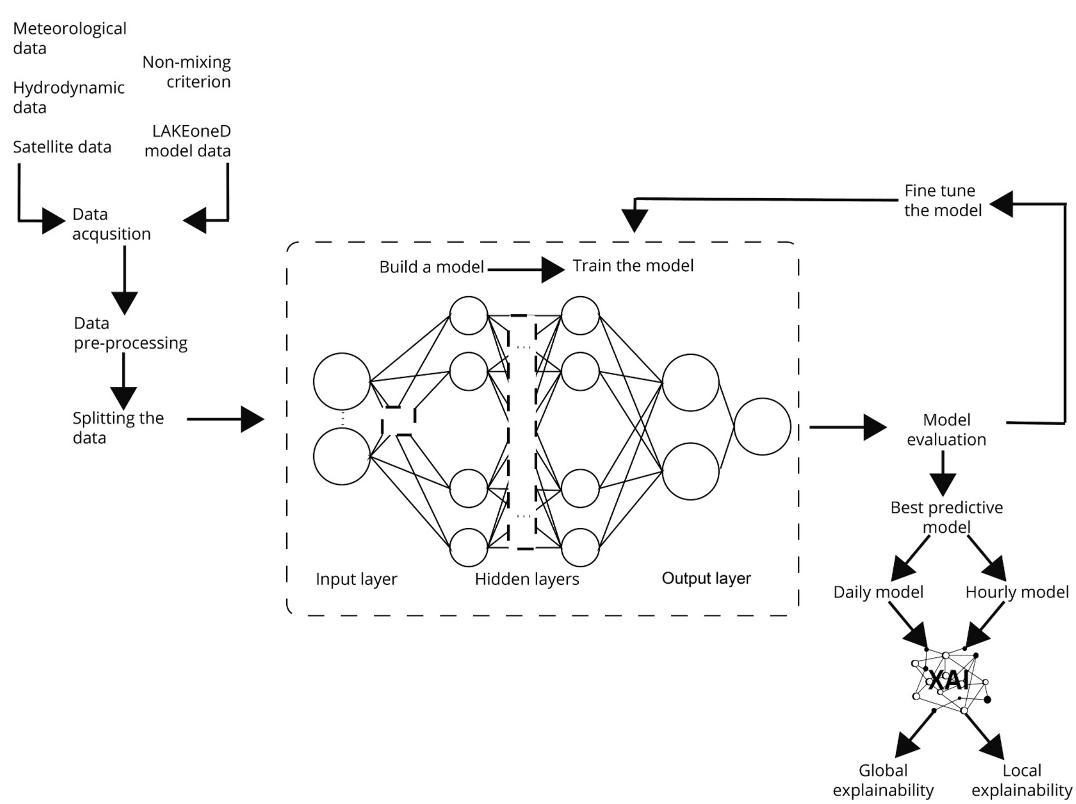


Fig. 3. Flowchart of the CNN-GRU model showing the main stages of the model development from data acquisition to the explainability stage.

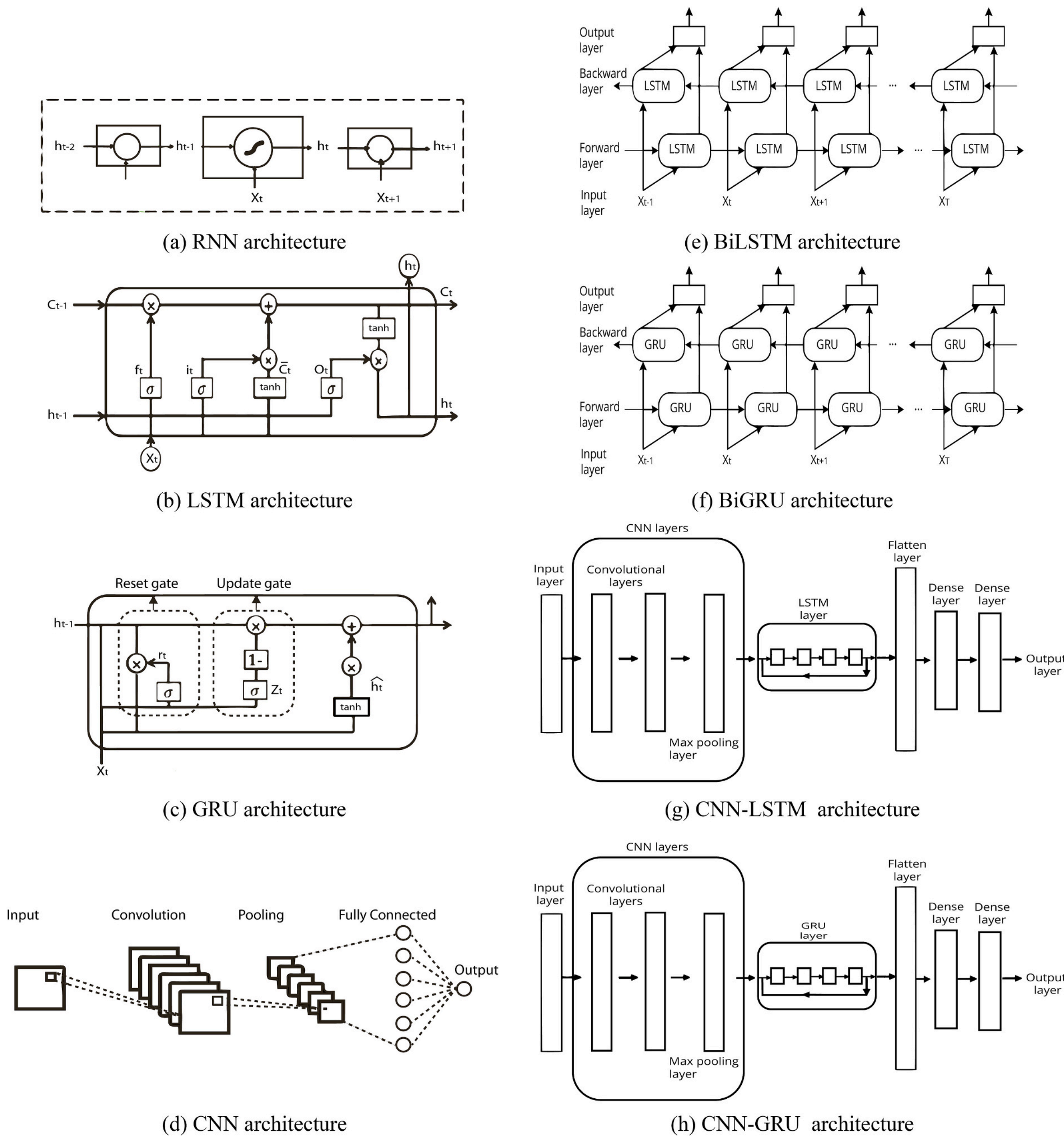


Fig. 4. Schematic diagrams representing a) RNN, b) LSTM, c) GRU, d) CNN, e) BiLSTM, f) BiGRU, g) CNN-LSTM and h) CNN-GRU models.

(Fig. 4f), and two hybrid models, Convolutional Neural Network-LSTM (CNN-LSTM) (Fig. 4g) and Convolutional Neural Network-GRU (CNN-GRU) (Fig. 4h). A hybrid model means two or more methods working together.

2.4.1. *Hyperparameter tuning*

Hyperparameter tuning involves systematically searching for the best hyperparameter values that maximise model performance. In this study, the Bayesian optimisation technique was employed to automate the tuning of algorithm-specific hyperparameters. Hyperparameters were tuned with Keras Tuner's Bayesian optimisation (num_initial_points = 50, max_trials = 15), minimising validation loss. Bayesian optimisation iteratively chooses hyperparameter settings to assess, guided by an acquisition function that balances exploration and exploitation. This strategy aims to find optimal hyperparameters with fewer evaluations compared to grid or random search (Joy et al., 2016).

Table 3 (a, b) presents the optimal architecture of both the proposed and benchmark models. Several key components significantly influence the design and functionality of neural networks. Filters play a crucial role in extracting complex patterns from input data, emphasising the importance of an efficient architecture to optimise this process. The arrangement and number of units or neurons (along with the layers they comprise) directly contribute to the network's capacity to learn and generalise from data. Furthermore, the choice of activation functions, such as ReLU, sigmoid, or tanh, is crucial for determining neurons' responsiveness to inputs, which ultimately affects the network's performance. The learning rate parameters must be carefully selected, as they significantly influence the convergence speed of the model towards minimising the weight values, thereby ensuring effective learning over iterations. Additionally, dropout techniques are applied as a regularisation strategy to enhance model robustness by mitigating overfitting. Concurrently, the kernel size used in convolutional layers dictates the

Table 3a
Optimal architecture of the proposed daily model and benchmark models.

Model	Hyperparameters	Search Space	Optimal hyperparameters
CNN-GRU	Filters	[64–320]	286
	Units	[32–128]	34
	Activation function	[“ReLU”,”tanh”,”sigmoid”]	ReLU
	Learning rate	[0.0001–0.002]	0.0014536
	Dropout rate	[0, 0.1, 0.2]	0.1
	Kernel size	[2,4]	
Simple RNN	Units	[32–64]	50
	Activation function	[“ReLU”,”tanh”,”sigmoid”]	ReLU
	Learning rate	[0.0001–0.002]	0.00119476
	Dropout rate	[0, 0.1, 0.2]	0.1
CNN-LSTM	Filters	[64–320]	126
	Units	[32–128]	34
	Activation function	[“ReLU”,”tanh”,”sigmoid”]	ReLU
	Learning rate	[0.0001–0.002]	0.0011424
	Dropout rate	[0, 0.1, 0.2]	0.2
	Kernel size	[2,4]	
BiGRU	Units	[32–64]	42
	Activation function	[“ReLU”,”tanh”,”sigmoid”]	ReLU
	Learning rate	[0.0001–0.002]	0.00192
	Dropout rate	[0, 0.1, 0.2]	0.1
BiLSTM	Units	[32–64]	36
	Activation function	[“ReLU”,”tanh”,”sigmoid”]	Tanh
	Learning rate	[0.0001–0.002]	0.0012015
	Dropout rate	[0, 0.1, 0.2]	0
LSTM	Units	[32–64]	44
	Activation function	[“ReLU”,”tanh”,”sigmoid”]	ReLU
	Learning rate	[0.0001–0.002]	0.00129485
	Dropout rate	[0, 0.1, 0.2]	0
GRU	Units	[32–64]	56
	Activation function	[“ReLU”,”tanh”,”sigmoid”]	ReLU
	Learning rate	[0.0001–0.002]	0.0010670
	Dropout rate	[0, 0.1, 0.2]	0

Table 3b
Optimal architecture of the proposed hourly model and benchmark models.

Model	Hyperparameters	Search Space	Optimal hyperparameters
CNN-GRU	Filters	[64–320]	282
	Units	[32–128]	34
	Activation function	[“ReLU”,”tanh”,”sigmoid”]	ReLU
	Learning rate	[0.0001–0.002]	0.0013889
	Dropout rate	[0, 0.1, 0.2]	0.2
	Kernel size	[2,4]	
CNN-LSTM	Filters	[64–320]	308
	Units	[32–128]	72
	Activation function	[“ReLU”,”tanh”,”sigmoid”]	Tanh
	Learning rate	[0.0001–0.002]	0.0004354
	Dropout rate	[0, 0.1, 0.2]	0.2
	Kernel size	[2,4]	
Simple RNN	Units	[32–64]	44
	Activation function	[“ReLU”,”tanh”,”sigmoid”]	ReLU
	Learning rate	[0.0001–0.002]	0.0014961
	Dropout rate	[0, 0.1, 0.2]	0
BiGRU	Units	[32–64]	56
	Activation function	[“ReLU”,”tanh”,”sigmoid”]	Tanh
	Learning rate	[0.0001–0.002]	0.0019786
	Dropout rate	[0, 0.1, 0.2]	0
BiLSTM	Units	[32–64]	60
	Activation function	[“ReLU”,”tanh”,”sigmoid”]	Tanh
	Learning rate	[0.0001–0.002]	0.0012288
	Dropout rate	[0, 0.1, 0.2]	0
LSTM	Units	[32–64]	64
	Activation function	[“ReLU”,”tanh”,”sigmoid”]	ReLU
	Learning rate	[0.0001–0.002]	0.0005901
	Dropout rate	[0, 0.1, 0.2]	0.2
GRU	Units	[32–64]	32
	Activation function	[“ReLU”,”tanh”,”sigmoid”]	ReLU
	Learning rate	[0.0001–0.002]	0.0007745
	Dropout rate	[0, 0.1, 0.2]	0.1

specificity and relevance of the features extracted.

2.5. Model performance

The performance metrics outlined in Eqs. (1)–(6) are used to evaluate the effectiveness, accuracy, and reliability of the proposed predictive models. They serve as essential indicators that help to understand how well these models capture the underlying data patterns and meet the intended objectives. In this study, models are evaluated using the following statistical metrics: Pearson's correlation coefficient (r), Root Mean Square Error (RMSE), Mean Absolute Error (MAE), Nash-Sutcliffe Efficiency Coefficient (NSE), modified Willmott's Index (WI), and Legate-McCabe Efficiency Index (LME). These metrics were calculated as follows:

$$r = \frac{\sum (y_i - \overline{y})(x_i - \overline{x})}{\sqrt{\sum (y_i - \overline{y})^2 \sum (x_i - \overline{x})^2}}, \tag{1}$$

$$RMSE = \sqrt{\frac{\sum (x_i - y_i)^2}{n}}, \tag{2}$$

$$MAE = \frac{|x_i - y_i|}{n}, \tag{3}$$

$$NSE = 1 - \frac{\sum_{i=1}^{n} (x_i - y_i)^2}{\sum_{i=1}^{n} (x_i - \overline{x})^2}, \tag{4}$$

$$WI = 1 - \frac{\sum_{i=1}^{n} |x_i - y_i|}{\sum_{i=1}^{n} (|y_i - \overline{x}| + |x_i - \overline{x}|)}, \tag{5}$$

$$LME = 1 - \frac{\sum_{i=1}^{n} |x_i - y_i|}{\sum_{i=1}^{n} |x_i - \overline{x}|}, \tag{6}$$

where x_i is the observed DO (mg/L), y_i is the predicted DO (mg/L), n is the total number of data points, $\overline{x}$ is the average observed DO (mg/L), and $\overline{y}$ is the average predicted DO (mg/L).

The coefficients r and NSE span the range from $-\infty$ to 1, with 1 representing an optimal value indicative of a perfect model. The coefficient r evaluates the degree of concordance between observed data and predicted values, while the NSE quantifies the model's capacity to predict values distinct from the observed mean of the target variable. However, it is important to note that the NSE may frequently overestimate larger values and underrepresent smaller ones. In contrast, the RMSE and MAE are constrained within the range of 0 to ∞, with 0 serving as the ideal value. These metrics assess the fit of the model but are often criticised for their sensitivity to outliers. Conversely, WI and LME also range from 0 to 1, with 1 being the optimal value. These indices present an advantage over r, RMSE, and MAE, as they do not attribute undue weight to outliers.

2.6. Explainability

In this study, we adopt the SHAP (SHapley Additive exPlanations) framework as a unified approach for interpreting model predictions, thereby addressing the challenge of understanding complex, black-box-style AI models. The SHAP approach assigns importance values to features for specific predictions, integrating six existing methods into a coherent class of additive feature attribution techniques. SHAP values are derived from game theory, ensuring a unique solution with desirable properties (Lundberg and Lee, 2017). The SHAP values are computed using:

$$\varphi(i) = \sum_{S \subseteq N/i} \frac{|S|!(|N| - |S| - 1)!}{|N|!} \left(\vartheta(S \cup \{i\}) - \vartheta(S)\right), \tag{7}$$

where S is an instance, $\vartheta(S)$ is the total value of S instances, and N is the number of instances.

We used the SHAP library in Python, version 0.42.1, to explain the model's predictions.

3. Results and discussion

3.1. Dissolved oxygen models

3.1.1. Daily model

The daily model operates on aggregated data collected over 24 h, providing a broader overview of processes and trends. It simplifies temporal resolution by focusing on daily maximums, minimums, or averages, thereby reducing data input requirements and computational complexity.

For each modelling experiment, a total of seven models were utilised. These include both the objective model (CNN-GRU) and its corresponding counterpart models (Simple RNN, CNN-LSTM, BiGRU, BiLSTM, LSTM and GRU).

Across all experiments (Tables 4 and 5), during both the training and testing periods, the CNN-GRU achieves the best performance in r (0.9040 and 0.9358), MAE (0.6544 and 0.6157 mg/L), RMSE (0.8913 and 0.7521 mg/L), NSE (0.8134 and 0.8648), WI (0.8066 and 0.8138), and LME (0.6168 and 0.6545). It captures the highest r, NSE, WI and LME and the lowest MAE and RMSE. Notably, the model performs better on the test dataset than on the training dataset, suggesting that the test set may contain fewer complex samples. Alternatively, since the data is split based on time, the test dataset may be more stable than the earlier period. So, the model's performance on the test dataset can be attributed to its exposure to weather conditions that were less extreme than those encountered during the training phase. By presenting patterns that had already been learned well during the model's learning process, the test data facilitated more accurate predictions.

Fig. 5 (a,b) illustrates the result of the daily time-step-based CNN-GRU model and its band plots. This is the test dataset, which corresponds to 25% of the total available data. The timeline spans from April 2023 to December 2024, with two peaks observed between July and August 2023 and 2024, and the lowest values occurring between December 2023 and March 2024. This figure illustrates the temporal progression and predictive accuracy of the daily model, demonstrating the model's ability to adapt to dynamic conditions. The model captures the variability in the data, even in periods of low DO values. We found that the DO generally peaked in winter and at the beginning of spring, then declined to its minimum in summer, and finally improved in autumn. DO concentrations generally display pronounced seasonal fluctuations, which are largely influenced by changes in water temperature, biological activity, hydrodynamics, and the input of organic matter (Mihuc et al., 1999).

Table 4
The training performance of the proposed daily model and the benchmark models.

Model	r	MAE (mg/L)	RMSE (mg/L)	NSE	WI	LME
CNN-GRU	**0.9040**	**0.6544**	**0.8913**	**0.8134**	**0.8066**	**0.6168**
Simple RNN	0.8871	0.6806	0.9636	0.7819	0.7936	0.5867
CNN-LSTM	0.8929	0.7349	1.0424	0.7448	0.7833	0.5624
BiGRU	0.8391	0.6785	1.1290	0.7006	0.7646	0.5495
BiLSTM	0.8373	0.7249	1.1315	0.6992	0.7580	0.5368
LSTM	0.8471	0.6967	1.1400	0.6947	0.7516	0.5338
GRU	0.8079	0.7181	1.2432	0.6370	0.7357	0.4921

Table 5
The testing performance of the proposed daily model and the benchmark models.

Model	r	MAE (mg/L)	RMSE (mg/L)	NSE	WI	LME
CNN-GRU	**0.9358**	**0.6157**	**0.7521**	**0.8648**	**0.8138**	**0.6545**
BiGRU	0.9268	0.6590	0.8345	0.8335	0.7998	0.6301
GRU	0.9057	0.7052	0.8855	0.8126	0.8126	0.6042
BiLSTM	0.9218	0.7067	0.8779	0.8158	0.8158	0.6033
LSTM	0.9097	0.7090	0.8868	0.8120	0.7792	0.6021
Simple RNN	0.9042	0.7223	0.8818	0.8141	0.7814	0.5946
CNN-LSTM	0.9294	0.7882	0.9629	0.7784	0.7621	0.5576

3.1.2. Hourly time-step model

The hourly time-step model is a predictive framework that operates on data aggregated at the hourly level.

All experimentally obtained modelling metrics (Tables 6 and 7) demonstrate that the objective model surpasses the competing models in predicting DO levels. Similar to the daily model, the best hourly model is the hybrid CNN-GRU, which achieves the highest r (0.9545 and 0.9234), NSE (0.9094 and 0.8487), WI (0.8642 and 0.8086), and LME (0.7329 and 0.6421), and the lowest MAE (0.4561 and 0.6391 mg/L) and RMSE (0.6339 and 0.6356 mg/L).

In this context, the CNN-GRU hybrid model is effective because the CNN layers capture local spatial/temporal patterns and reduce input dimensionality (Yamashita et al., 2018), while the GRU units model longer-term temporal dependencies with fewer parameters and lower computational cost than alternative recurrent (Cho et al., 2014). In preliminary experiments (Tables 4 to 7), other hybrid combinations (such as CNN-LSTM) either showed slightly inferior performance while requiring more complex training and higher computational resources. These findings suggest that this hybrid approach demonstrates significant advantages in processing temporal data.

Fig. 6 (a, b) presents the result of the hourly CNN-GRU model and its band plots. The temporal scope extends from April 2023 to December 2024. This cyclical pattern results in characteristic diurnal oscillations that can vary in amplitude depending on factors such as temperature and biological activity.

First, elevated temperatures diminish the solubility of oxygen in water, resulting in decreased DO concentrations during warmer periods. Conversely, cooler temperatures at night tend to increase oxygen solubility, facilitating oxygen re-aeration. Also, stratification breakdown leads to diluted DO throughout the water column during summer nights. Some studies have documented that temperature fluctuations are a primary driver of diurnal oxygen variation, with the amplitude of these fluctuations often correlating with the extent of temperature change (He et al., 2022). Second, the primary biological processes influencing diurnal oxygen variation in rivers are photosynthesis and respiration. During daylight hours, aquatic plants, algae, and phytoplankton perform photosynthesis, producing oxygen and elevating DO levels in the water column (Uchrin et al., 1988). On the contrary, respiration by aquatic organisms and microbial communities consumes oxygen continuously, with heightened activity often occurring at night when photosynthesis ceases (Williams et al., 2000).

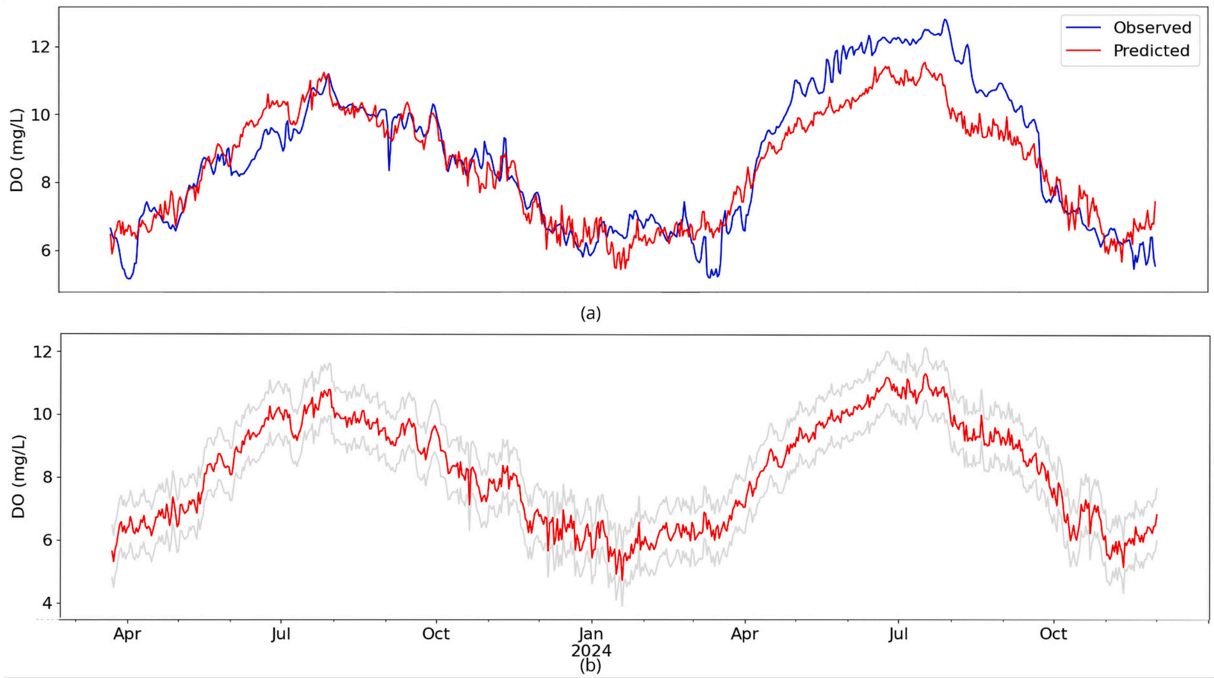


Fig. 5. Daily dissolved oxygen (DO) (mg/L) predicted by the CNN-GRU model at Redbank Weir during the testing period, a) capturing the seasonal pattern, compared with the observed DO daily timeseries, b) including error bands with the predicted DO daily time series.

Table 6
The training performance of the proposed hourly model and the benchmark models.

Model	r	MAE (mg/L)	RMSE (mg/L)	NSE	WI	LME
CNN-GRU	**0.9545**	**0.4561**	**0.6339**	**0.9094**	**0.8642**	**0.7329**
CNN-LSTM	0.9403	0.5685	0.7235	0.8820	0.8413	0.6961
Simple RNN	0.9370	0.5084	0.7447	0.8750	0.8387	0.6891
BiGRU	0.9347	0.4972	0.7550	0.8752	0.8376	0.6848
BiLSTM	0.9170	0.5767	0.8482	0.8378	0.8209	0.6536
LSTM	0.9033	0.5437	0.9266	0.8064	0.7968	0.6196
GRU	0.8883	0.5738	0.9866	0.7806	0.7840	0.5934

Table 7
The testing performance of the proposed hourly model and the benchmark models.

Model	r	MAE (mg/L)	RMSE (mg/L)	NSE	WI	LME
CNN-GRU	**0.9234**	**0.6391**	**0.6356**	**0.8487**	**0.8086**	**0.6421**
CNN-LSTM	0.9205	0.6571	0.8310	0.8357	0.7999	0.6320
BiGRU	0.9186	0.6645	0.8312	0.8356	0.7959	0.6276
Simple RNN	0.9122	0.6865	0.8439	0.8305	0.7949	0.6155
BiLSTM	0.9196	0.6947	0.8549	0.8261	0.7831	0.6109
GRU	0.9302	0.7294	0.8711	0.8194	0.7672	0.5915
LSTM	0.9198	0.7539	0.8966	0.8087	0.7581	0.5778

The daily model was constructed utilising a set of 19 features, integrating the mean, minimum, and maximum values of each feature. In contrast, the hourly model utilised 10 features, derived from the mean hourly values of these variables. Both models handle seasonal trends, but the hourly model shows greater accuracy in capturing fine-grained daily patterns. So, hourly models can surpass when system behaviour varies significantly across hours. Accurate prediction of hourly DO levels is vital for effective water quality management, enabling timely interventions to prevent hypoxic conditions and maintain ecological balance. However, one of the primary challenges in accurately predicting hourly DO levels is the quality and availability of data.

Using the Mann–Kendall test with Sen's slope, we found that dissolved oxygen in the Murrumbidgee River shows a very small trend per year (−0.00033 mg/L per year), from 2017 to 2024, which remained of similar magnitude under sensitivity analyses (exclusion of extreme years). Also, model performance metrics (such as r, NSE, WI, LME) against variations in input parameters like learning rate achieve the peak performance at a moderate value (learning rate of 0.001), while lower values cause underfitting and higher values introduce instability. In summary, the trend analysis confirms a consistent directional change, and the performance metrics demonstrate the extent to which this finding holds across parameter changes.

3.2. *Explainability*

3.2.1. *Global explainability*

Global explanations provide insights into the overall behaviour of the model across all predictions. Among the various visualisation techniques available in SHAP, the beeswarm plot stands out as an effective method. In the beeswarm plot, each point represents a SHAP value for a feature and an instance, allowing for a visual representation of the distribution of feature impacts across all instances. So, this visualisation not only shows the relative importance of features but also illustrates the direction of their impact.

In this plot, features are listed on the y-axis and sorted by importance. The higher the position, the higher the importance. For each variable, each instance in the dataset is represented as a point, and these points are distributed along the x-axis according to their SHAP values. Those values represent how these inputs feature impact the final prediction. Those values have two components: magnitude and direction. The higher the magnitude, the greater the impact of this feature on the prediction; some values are negative, while others are positive. The colour bar corresponds to the raw values of the variables for each

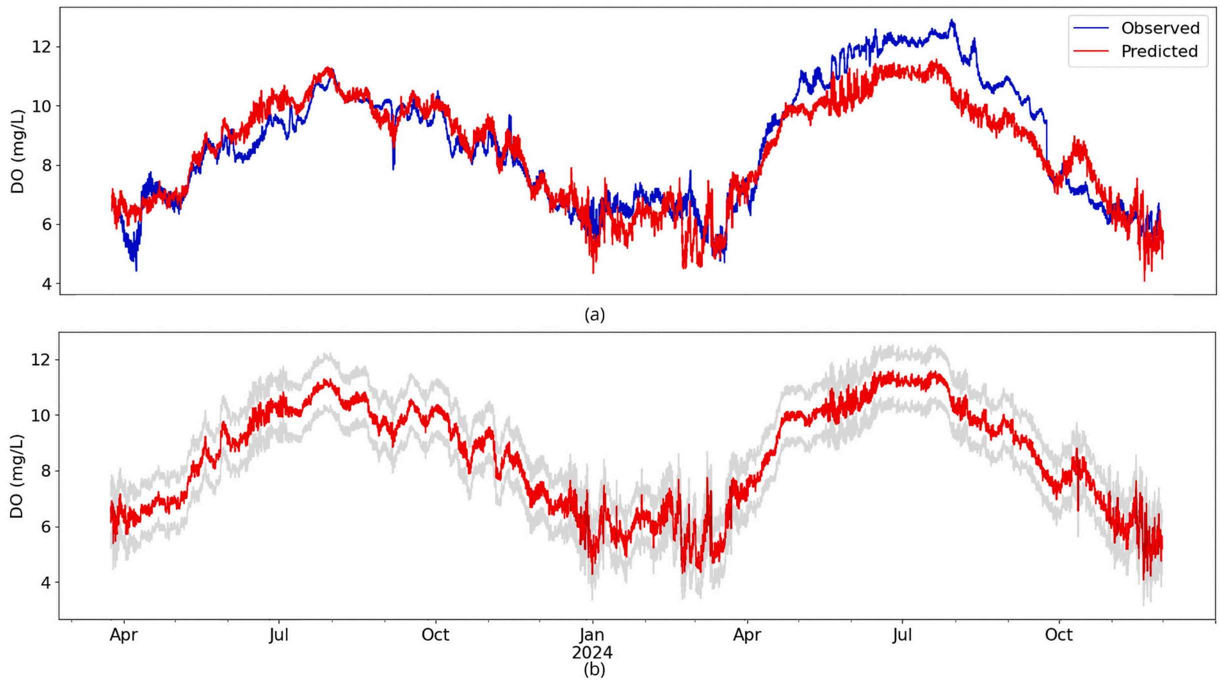
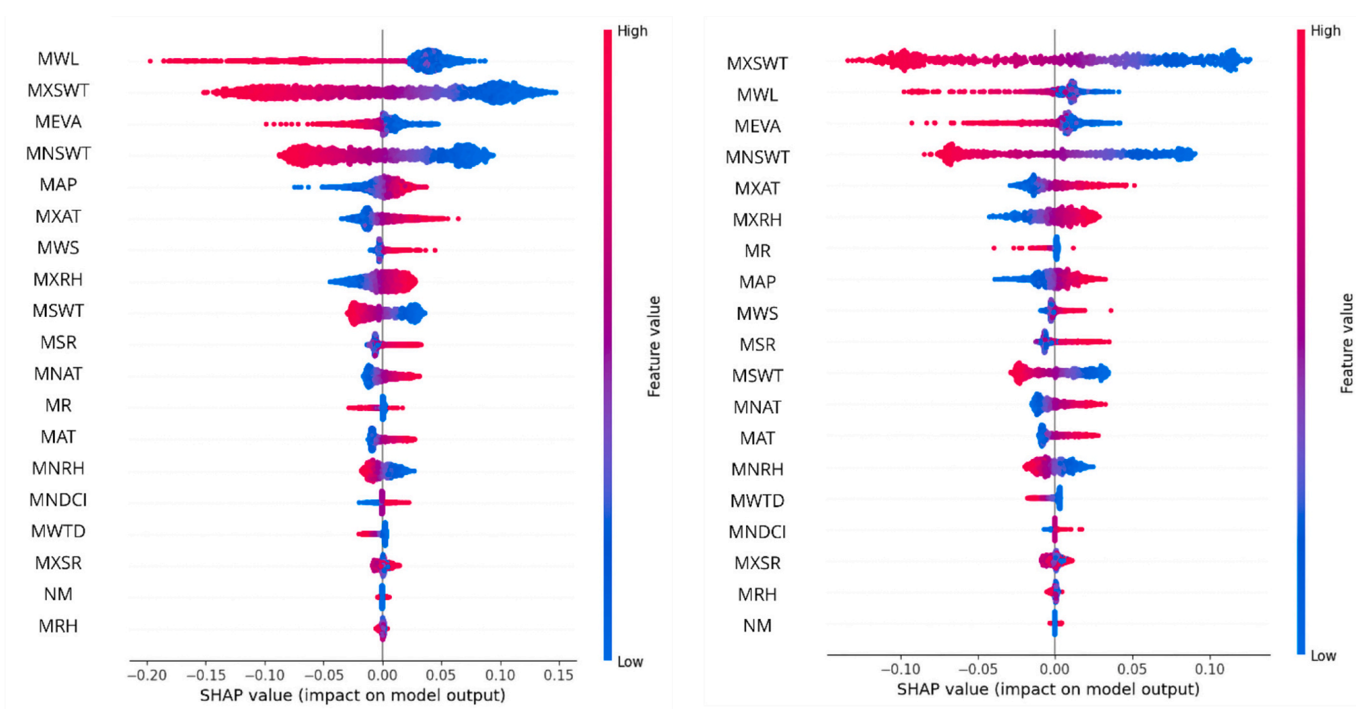


Fig. 6. Hourly dissolved oxygen (DO) (mg/L) predicted by the CNN-GRU model at Redbank Weir during the testing period, a) capturing the seasonal and daily patterns, compared with the observed DO hourly time series, b) including error bands with the predicted DO hourly time series.

instance on the plot.

3.2.1.1. Global explainability for the daily model. Fig. 7 displays the SHAP beeswarm plot for the training and testing period. The MWL is the most important global feature at the top (Fig. 7a) during the training period for the daily model. For the water level, we observe a dense cluster of instances with low water levels (blue points) and positive SHAP values, indicating that this feature has a similar impact on the model's output. Instances of higher water levels (red points) extend from the right to the left, indicating greater variability in how that feature influences predictions across samples. Suggesting that high water level values often have a stronger negative impact on DO; however, there are instances where high water levels have a positive impact on DO as well.

In comparison, low water level values are associated with a positive



(a) (b)

Fig. 7. Global explainability, beeswarm plot for the daily CNN-GRU model at Redbank Weir, a) training period, b) testing period. See nomenclature in Table 1.

impact on DO prediction. Fluctuations in water level can influence the extent of oxygen exchange between water and the atmosphere, as well as affect the physical and biological processes that regulate oxygen dynamics. For instance, lower water levels often result in reduced surface area for gas exchange, leading to decreased oxygen diffusion into the water column (Regmi et al., 2021). Conversely, higher water levels can enhance aeration and replenish oxygen, especially in shallow systems where wind and wave action facilitate oxygen transfer (Baxa et al., 2021). The group of low MWL values that positively impacts the DO prediction could be associated with two natural sources: atmospheric diffusion and photosynthesis by aquatic plants and phytoplankton.

MXWT emerged as the second most influential global feature during the training period. Lower MXWT values are generally associated with a positive impact on DO concentrations, likely due to the greater solubility of oxygen in cooler water and reduced metabolic consumption by aquatic organisms. In contrast, higher MXWT values tend to negatively affect DO, as warmer water holds less oxygen and elevated temperatures increase the respiration rates of aquatic biota, thereby accelerating oxygen depletion. This relationship highlights the critical role of thermal dynamics in regulating DO variability and validates the importance of incorporating MXWT as a key predictor in predictive models of aquatic oxygen levels.

MEVA is the third most important global feature, and similarly, lower evaporation rates have a positive impact on DO.

The fourth most important global feature is MNSWT. Lower MNSWT values are generally associated with a positive influence on DO predictions, likely because cooler water can hold more oxygen and reduces metabolic consumption by aquatic organisms. Conversely, higher MNSWT values negatively impact DO predictions, as elevated temperatures decrease oxygen solubility and can accelerate biological oxygen demand. This relationship underscores the critical role of thermal conditions, particularly during periods of minimum temperature, in regulating DO variability and highlights the importance of incorporating MNSWT as a key predictor in aquatic oxygen models.

Mean air pressure level ranks as the fifth most important global feature affecting DO. Lower air pressure values tend to negatively impact DO, potentially due to reduced oxygen solubility and enhanced gas exchange dynamics, whereas higher air pressure values are associated with positive impacts on DO, likely facilitating greater oxygen dissolution at the water surface. This relationship highlights the significant role of atmospheric pressure in modulating DO variability and emphasises the need to include mean air pressure as a key predictor in models of aquatic oxygen levels.

Two important features to mention are the MNDCI and the NM. MNDCI shows a direct relationship: higher values have a positive impact on DO, while lower values have a negative impact. NM, on the contrary, does not present a particular pattern in the global explainability of this period, with a high concentration of dots around SHAP value = 0.

During the testing period (Fig. 7b), similar to the training period, MXSWT, MWL, MEVAP, and MNSWT are the four most important global features. However, during this period, MXSWT plays a more significant role than it did during the training period. During the testing period, MNDCI and NM are in lower positions than during the training period, suggesting their impact on the target variable is lower. However, they exhibit behaviour similar to that during the training period.

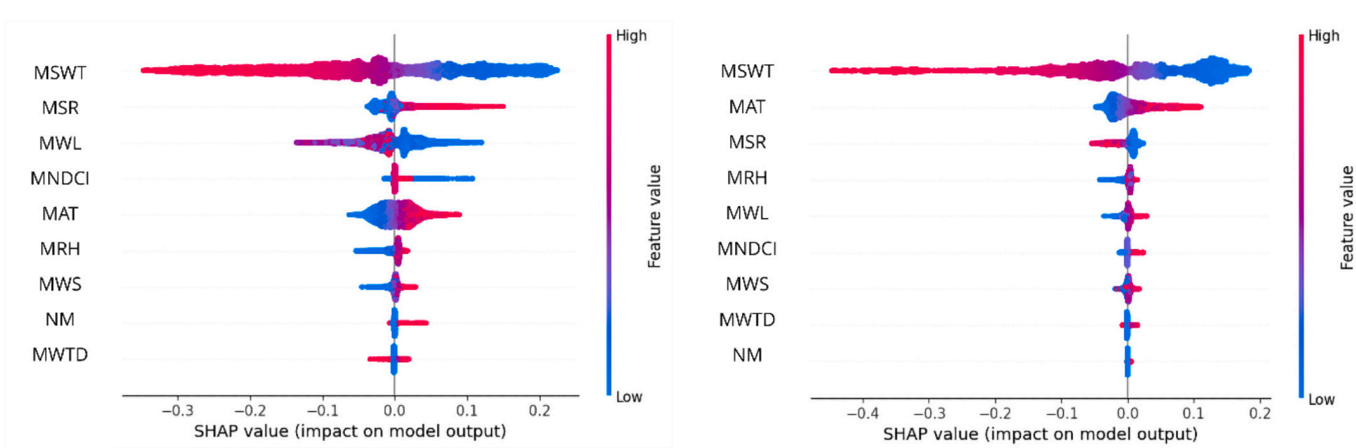
In summary, SWT is the primary predictor, MWL the secondary predictor and EVA the tertiary predictor.

3.2.1.2. Global explainability for the hourly model. Fig. 8 shows beeswarm plots for the hourly model, Fig. 8a for the training period and 8b for the testing period. The MSWT is the most important variable in both the training and testing periods, with lower values having a positive impact and higher values having a negative impact on DO predictions.

During the training period, the second position is occupied by MSR; during the testing period, it is occupied by MAT. A study in the Murray-Darling Basin reported that air temperature is the most important predictor of surface water temperature (Briceno Medina et al., 2024), which is highly associated with other variables. In addition, air temperature is strongly connected to both the stratification and destratification processes occurring within weir pools (Bormans and Webster, 1997, 1998).

The MNDCI shows a higher position (fourth) in the hourly training period than in the testing period. NM is almost always in the last position during the training period, exhibiting a pattern of generally higher values that positively affect the SHAP values; however, there are a few instances where higher values negatively affect the SHAP values. This suggests that, despite the existence of stratification, the upper layer generally remains well-oxygenated. Nevertheless, occasional low DO values are detected, possibly associated with destratification and mixing that can reduce oxygen levels. DO is measured at the surface, and the surface layer remains better oxygenated than the lower layers. For this reason, lower layers may have low oxygen levels, while the upper layers still have normal oxygen levels. Therefore, only when destratification and mixing occur, low DO values are reflected in the surface layer. As such, monitoring DO levels at the surface alone may not provide a comprehensive picture of the aquatic environment's condition.

Consequently, in the hourly case, as in the daily case, the MSWT plays a very important role in predicting DO during the training and testing periods. Water temperature has a well-documented inverse relationship with DO levels, primarily because temperature affects oxygen solubility in water. As temperature increases, the solubility of



(a) (b)

Fig. 8. Global explainability, beeswarm plot for the hourly CNN-GRU model at Redbank Weir, a) training period, b) testing period. See nomenclature in Table 1.

oxygen decreases, leading to lower DO concentrations in the water column (Hussain et al., 2010; Zhong et al., 2021). The relationship between these two variables is complex and is influenced by seasonal variations, biological activity, and anthropogenic factors such as pollution and climate change (Jane et al., 2021).

Summarising, SWT is the primary predictor, and other predictors vary according to the training and testing period.

3.2.2. Local explainability

Local explanations pertain to the interpretation of individual predictions, providing insight into why a model generated a specific output for a particular instance. This approach enables a detailed understanding of the model's decision-making process at the level of single observations.

Among the visualisation methods offered by SHAP, the local bar plot provides a graphical representation of each feature's contribution to the specific prediction. In this plot, each bar represents a feature, with its length indicating the SHAP value for that feature in the particular prediction under consideration. Red bars extending to the right represent positive SHAP values, indicating features that increase the predicted outcome. While blue bars extending to the left represent negative SHAP values, they signify features that contribute to decreasing the prediction. The length indicates the magnitude of the SHAP value and the strength of the feature's influence on the prediction.

3.2.2.1. Local explainability for the daily model. Fig. 9 shows local bar plots for the daily model. These plots correspond to previous days (Fig. 9a and b) and to days during the fish kill event on the 26th and 27th of January 2019 (Fig. 9c and d). To analyse this event, the final model was applied to the 2018–2019 dataset for explainability analysis.

Before and during the fish kill event for daily predictions, surface water temperature (MXSWT and MNSWT) and MWL are the most important features. Water temperature lowers the prediction (MXSWT, MNSWT, and MSWT), while MWL increases it. During January 2019, in the lower Murrumbidgee catchment between Hay and Balranald, river levels and flows were low, and water temperatures were relatively warm, ranging from 26 to 27 °C (Murray–Darling Basin Authority, 2019).

Other features that drive the prediction lower, previous days of the event, are MXRH, MEVA, MAP, NM, and MNDCI. During the event, features that contribute to a reduction in the predicted values are MXRH, MEVA, MAP, MSR and MNDCI.

MEVA is in fourth position, days prior to the fish kill event, which lowers the prediction. However, during the fish kill event, it moved to position eight on the 26th of January 2019, and then to position 13 on the 27th of January 2019.

In short, SWT is the primary predictor and then appears MWL.

Before the 2019 event, the average SHAP value of SWT was higher than its value during the event. Over the two days of the event, SWT and WL remained in the same positions, with values remaining almost constant. This indicates that the importance or effect of the corresponding features on the model's predictions decreased across those periods. Thus, the model was more sensitive to these features before the event, while during the event, the marginal contribution of those features was stable. These features also ranked highly in the overall explainability, demonstrating their contribution before, during, and after the event.

During the event, NM goes from pushing the prediction lower to higher, which may be associated with destratification. On the contrary, the MNDCI continues to push the prediction lower during the 26th and 27th January 2019. Even when NM and MNDCI have a lower impact on DO prediction, NM is more important than MNDCI in the days preceding the fish kill event. Conversely, during the event, MNDCI is more important for predicting DO than NM.

In January 2019, a blue-green algal bloom occurred in the Murrumbidgee River. During that period, blue-green algae levels in the Redbank Weir pool were at amber alert thresholds. In December 2018, elevated concentrations of blue-green algae were detected at Hay, Maude, and Redbank Weirs; however, no fish deaths were reported at these locations. Although the algal bloom did not directly affect the fish population, its influence on surface oxygen levels may have indirectly affected the fish by exacerbating oxygen stress (Baldwin, 2019). Unlike Redbank, other weir pools in the Murrumbidgee River did not experience fish kills, highlighting the unique conditions at Redbank Weir during this event.

It is important to note the limitations of post-hoc explainability methods. SHAP and related techniques attribute model predictions to input features in a correlational sense and evaluate the contribution of various input variables (Alizamir et al., 2024); they reveal which variables the model relies on, but they do not establish causal relationships or mechanisms in the underlying physical system (Ng et al., 2025).

3.2.2.2. Local explainability for the hourly model. Figs. 10a to 10d exhibit four hours of the 25th of January 2019, which is a day before the fish kill event. For the analysis of this event, the 2018–2019 dataset was used to apply the final model for explainability purposes. Similar to the daily case, the most important feature is MSWT. First, at 9 am, the most important features are water temperature, water level, and solar radiation. Second, at 12 pm, the more important features are: surface water temperature, solar radiation, and relative humidity. Third, at 3 pm, the more important features are: surface water temperature, relative humidity, and air temperature. Finally, at 6 pm, the more important features are: surface water temperature, air temperature, and relative humidity.

Features that lower the prediction during the whole day are MSWT, MRH, and MNDCI. The features that positively influence the prediction throughout the entire day are MAT, MSR, and NM. NM exhibits greater significance at 6 pm compared to other times of the day. Overall, these temporal variations underscore the dynamic contributions of various environmental factors to the model's predictions throughout the daily cycle.

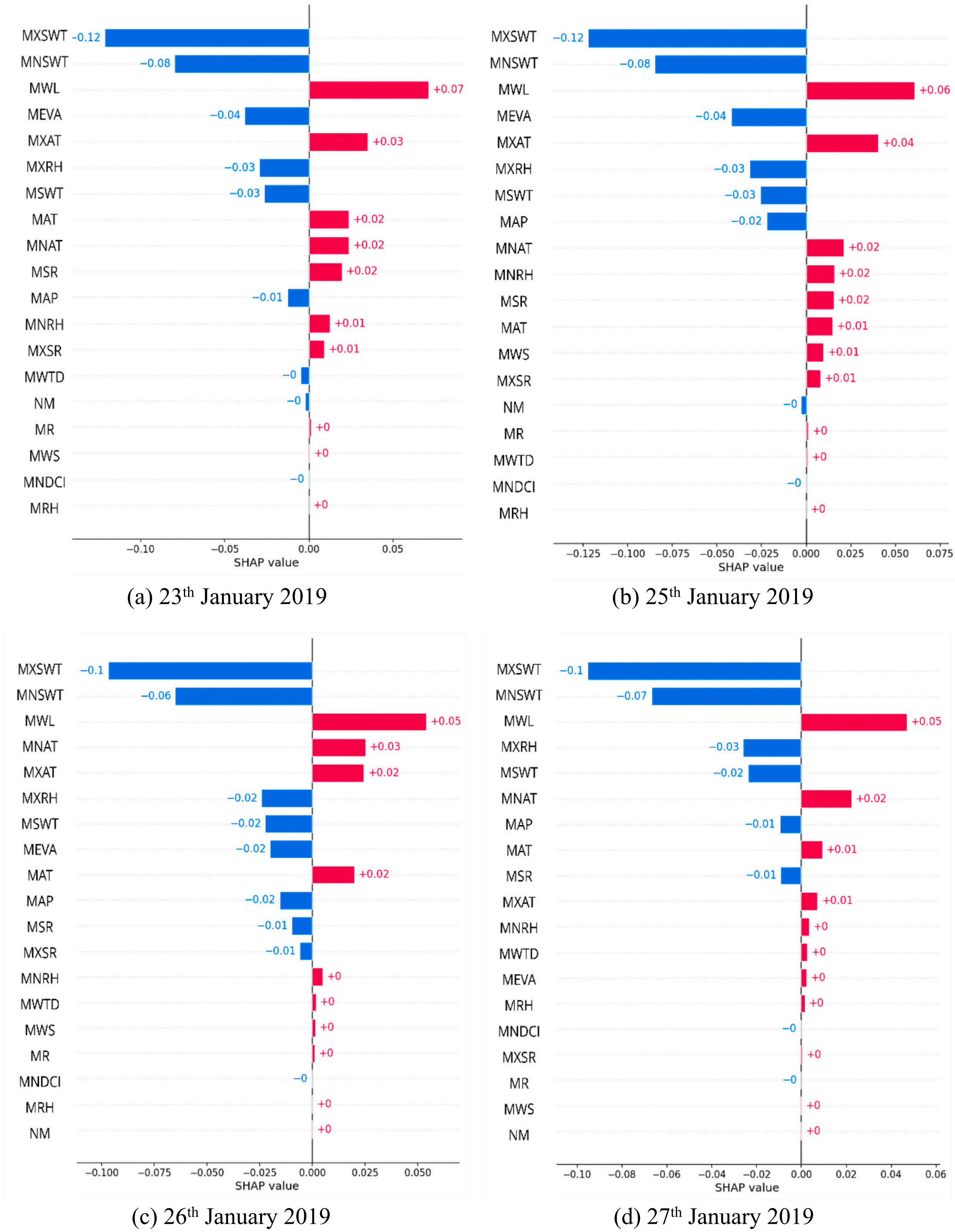
At the time of the fish kill event, Redbank Weir was nearly at full capacity, and the average DO concentration at a depth of below 2 m dropped from 6.2 mg/L on the 21st of January to 2.4 mg/L on the 29th of January (Baldwin, 2019).

Summarising, SWT is the primary predictor and other predictors vary according to the hour.

Figs. 11a to 11d present four hours on the 26th of January 2019, which is the first day of the fish kill event studied. On this day, as on the previous day, the most important feature is the MSWT. Thus, at 9 am, 12 pm, 3 pm, and 6 pm, the most important features are surface water temperature, air temperature, water level, and relative humidity.

NM appears slightly more important than the MNDCI during these hours. In addition, both NM and MNDCI negatively impact the DO prediction throughout the day. On the contrary, air temperature contributes positively, enhancing the predicted DO levels during these same hours. It highlights the complex interplay between physical and biological factors in regulating DO dynamics.

According to Baldwin (2019), the fish kill at Redbank Weir on January 26th – 27th coincided with a similar event in the Darling River at Menindee, both of which were linked to the same cold front. In addition, another factor that could have contributed to the fish kill at Redbank Weir was the extreme air temperatures in January 2019, with maximum daily temperatures of 35 °C or higher recorded at Balranald. Thus, in the study, it was stated that the fish kill at Redbank Weir on January 26–27, 2019, was primarily caused by the mixing of hypoxic bottom waters with well-oxygenated surface waters due to a cool change. This destratification resulted in insufficient DO levels throughout the water column, leading to the death of several thousand fish. Additionally, this event coincides with the release of water from



(a) 23th January 2019

(b) 25th January 2019

(c) 26th January 2019

(d) 27th January 2019

Fig. 9. Local explainability, bar plot for the daily CNN-GRU model at Redbank Weir, January 23–27 a) the 23rd of January 2019 b) the 25th of January 2019 c) the 26th of January 2019 d) the 27th of January 2019. See nomenclature in Table 1.

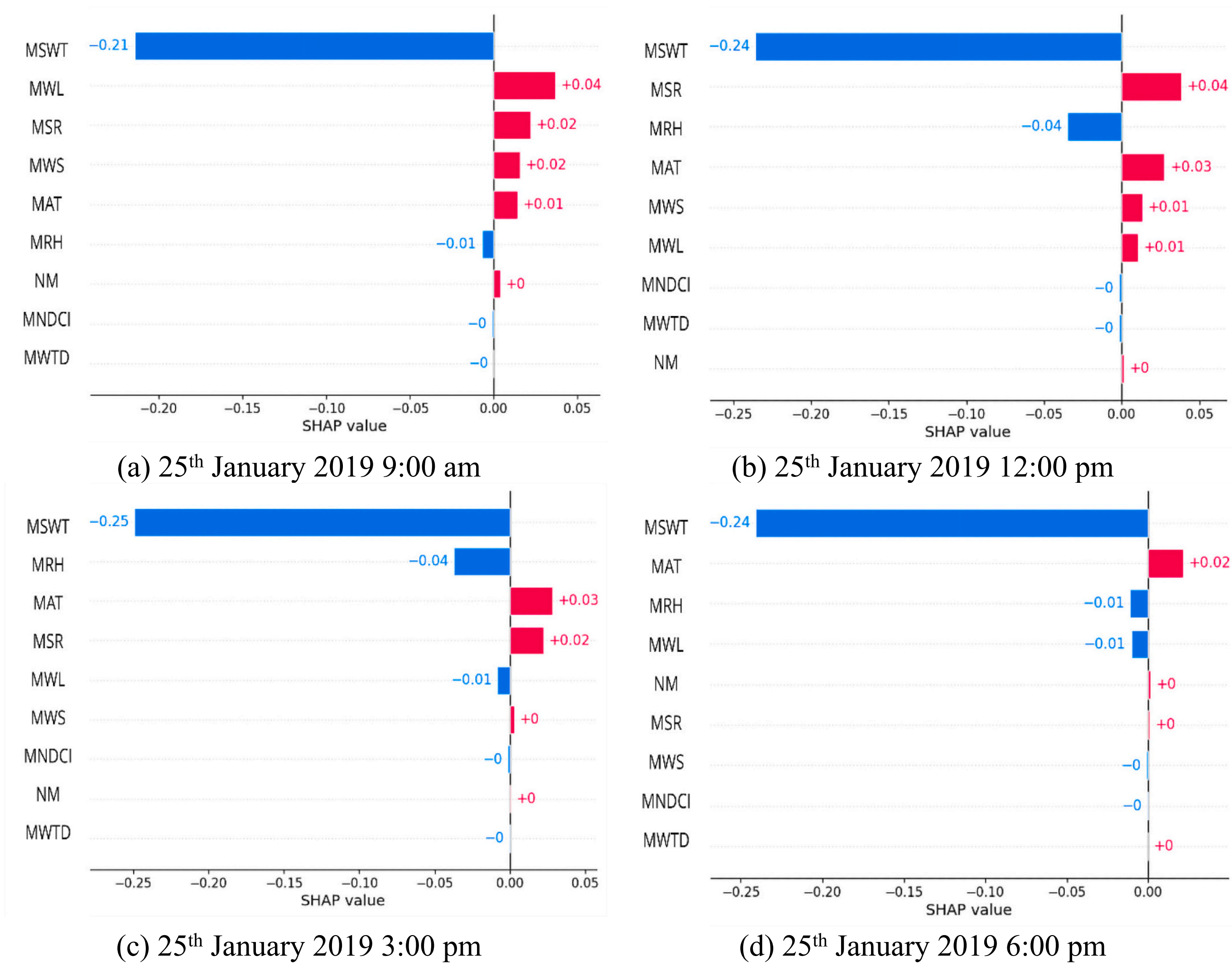


Fig. 10. Local explainability bar plot for the hourly CNN-GRU model at Redbank Weir, January 25 a) the 25th of January 2019 9:00 am b) the 25th of January 2019 12:00 pm c) the 25th of January 2019 3:00 pm d) the 25th of January 2019 6:00 pm. See nomenclature in Table 1.

Maude Weir to increase flow over Balranald Weir, which then passes through Redbank Weir.

It highlights that spatially and temporally linked factors can exacerbate ecological crises. Moreover, the role of extreme air temperatures and water management practices, such as the release from Maude Weir, must be meticulously considered, as they could contribute to conditions conducive to hypoxia. Therefore, identifying interactions between hydrodynamic changes and biotic responses will be crucial for developing effective strategies to mitigate the impacts of future fish kills. Integrating high-resolution monitoring data with predictive modelling can provide early warnings and inform adaptive management decisions. Additionally, fostering a comprehensive understanding of both environmental drivers and ecosystem responses will be essential for sustaining aquatic health under changing climatic and anthropogenic pressures.

In short, SWT is the primary predictor, and MWL appear as the secondary predictor.

In the study context, low DO during fish kills was primarily driven by changes in water temperature, as corroborated by the global and local explainability. As a consequence, extreme water temperatures and low-level conditions exacerbate oxygen depletion.

A limitation of this study is that the applied models did not explicitly address confounding interactions between explanatory variables or spatiotemporal dependencies present in the data. These factors may have influenced the stability of feature importance estimates and the interpretability of model outputs, potentially affecting the robustness and generalizability of the findings. However, we acknowledge that future work could benefit from spatial-temporal modelling to account for the identified complexities, and more extensive diagnostic and statistical validation should be undertaken.

Additionally, in this study, we identify both aleatory and epistemic uncertainties. Aleatory uncertainty arises within time series due to observational noise, which can lead the model to misattribute random fluctuations as indicative of underlying trends or seasonal patterns. On the other hand, epistemic uncertainty occurs when the available data from the physical-based model is limited, biased, or unrepresentative, resulting in significant uncertainty regarding model parameters. Consequently, the learned trend can change markedly if the training set is perturbed. Thus, future work should incorporate sensitivity and reliability analyses that decompose total output uncertainty into contributions from each epistemic input source, supporting ranking and targeted data collection or model refinement.

Also, it is important to acknowledge that extreme years may disproportionately influence the observed patterns. Given the considerable variability observed across years—particularly those characterised by extreme climatic conditions—further analyses should examine the extent to which these outlier years affect the robustness of the results. Sensitivity testing will be essential to confirm that the main conclusions hold across alternative temporal subsets.

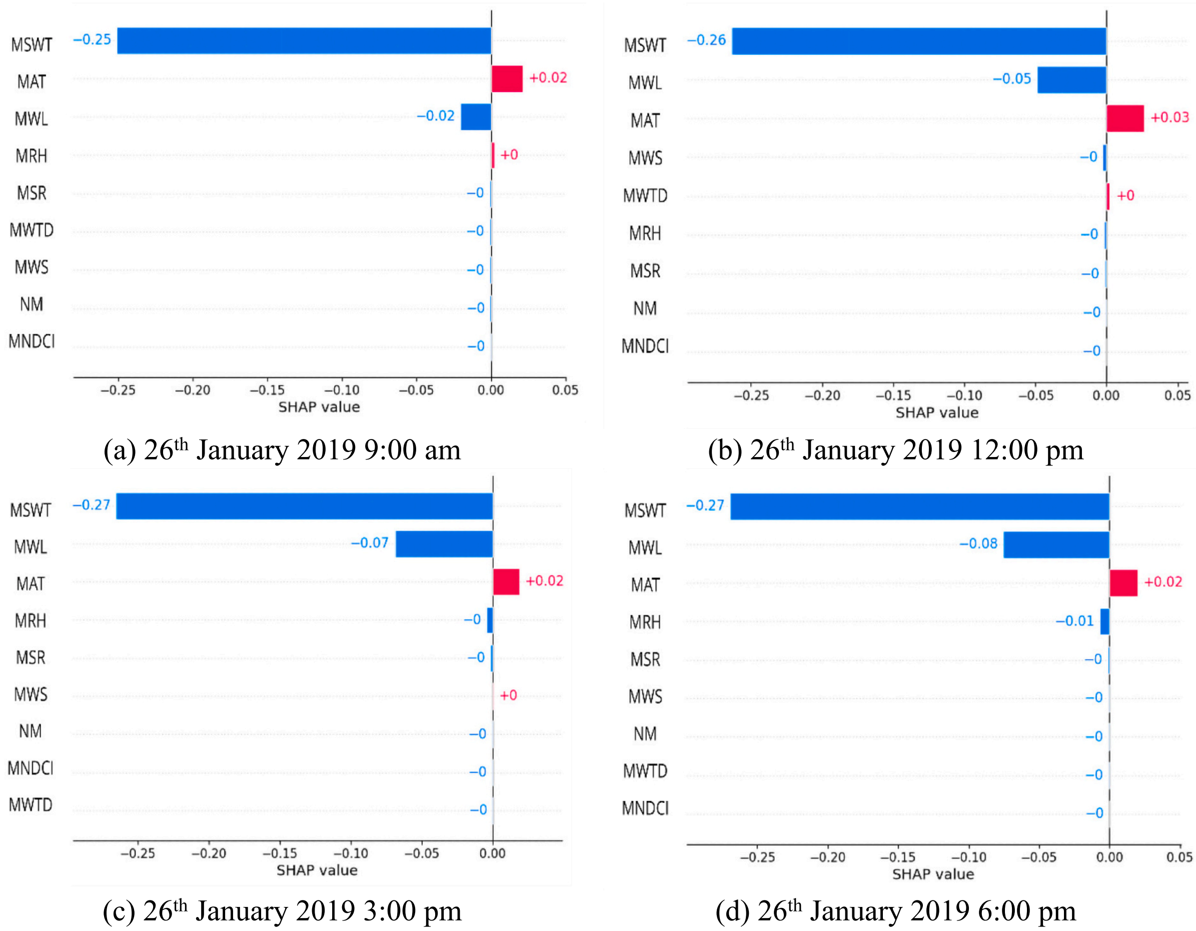


Fig. 11. Local explainability bar plot for the hourly CNN-GRU model at Redbank Weir, January 26 a) the 26th of January 2019 9:00 am b) the 26th of January 2019 12:00 pm c) the 26th of January 2019 3:00 pm d) the 26th of January 2019 6:00 pm. See nomenclature in Table 1.

4. Conclusions

This present study developed xAI models to predict daily and hourly DO levels in a case study at the Murrumbidgee River in the Murray-Darling Basin. The proposed predictive model, CNN-GRU, uses explainability to build trust with end users. The results verified that the daily and hourly CNN-GRU models significantly outperform the benchmark models in prediction. During both the training and testing phases, the CNN-GRU model outperforms others in daily and hourly predictions, demonstrating superior performance across metrics such as r, MAE, RMSE, NSE, WI, and LME.

The analysis presented the significance of complex interactions among environmental variables associated with low DO levels using explainable artificial intelligence methods. It underscores the critical role of surface water temperature as the predominant factor influencing DO predictions in both daily and hourly models. While water level and evaporation demonstrate their importance, surface temperature's consistent significance throughout the training and testing phases reinforces its centrality in aquatic health assessments. As corroborated by the global and local explainability, fluctuations in temperature significantly influence oxygen depletion, creating conditions that can lead to devastating impacts on aquatic ecosystems. This suggests that monitoring water temperature is vital in predicting and preventing fish kills. Overall, these insights stress the importance of understanding environmental factors and their interplay in maintaining the health of aquatic ecosystems.

However, since this study is data-driven, missing information within the dataset could negatively affect the accuracy of the model. So, missing data is a limitation in the data collection process, especially for satellite data used to estimate the MNDCI, which affects the model's accuracy. Furthermore, the non-mixing is based on a simplified definition, which also introduces bias to the model's accuracy. This emphasises the necessity that future implementations prioritise higher data acquisition frequency and more sophisticated modelling techniques to improve temporal resolution and operational responsiveness.

To facilitate a more comprehensive understanding, future investigations must integrate additional variables, including sedimentation and salinity, alongside environmental fluctuations associated with climate change. This approach will likely yield insights into the interactive effects of these parameters on DO levels.

Furthermore, this research was carried out as a case study, extracting data only from the Redbank Weir in the Murrumbidgee River, extending the application of this model to various locations within the Murray-Darling basin will not only enhance its robustness but also determine the feasibility of generalisations across different ecosystems.

Overall, the findings of this study not only contribute to understanding the specific events surrounding the fish kill at Redbank Weir but also encourage broader consideration of ecological management in the context of changing climate patterns and water resource use. In addition, integrating explainable methods with deep learning models represents an advancement in this field. Then, such models could be incorporated into real-time water management systems for early warning of low DO conditions. By prioritising the development of such models, we can enhance our understanding of environmental dynamics. Therefore, this study provides an explainable deep learning framework that could be adaptable to other aquatic systems, such as rivers, shallow lakes and reservoirs, across semi-arid regions, ultimately contributing to more effective environmental monitoring and management strategies.

CRediT authorship contribution statement

Leyde Briceno Medina: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Duy Nguyen:** Writing – review & editing, Visualization, Validation, Methodology, Data curation, Conceptualization. **Klaus Joehnk:** Supervision, Resources, Funding acquisition, Conceptualization. **Yiqing Guo:** Writing – review & editing, Validation, Software, Methodology, Data curation. **Ravinesh C. Deo:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition. **Salvin Prasad:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data used in this study were obtained from various sources. These have been mentioned in Section 2.2. The dataset in its current form, along with the Python code are available via the following GitHub link: https://github.com/DORBW/DO

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